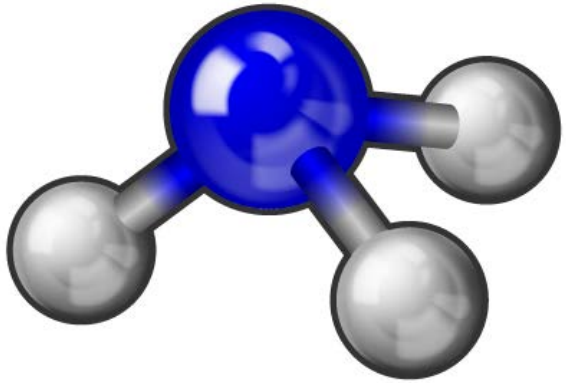
“A” IS FOR AMMONIA: Why and how fertilizer may power the planet

Nicholas E. Thornburg, Ph.D.

Metro Denver CRES Seminar

January 16, 2025



Senior Chemical Reaction Engineer
Energy Conversion and Storage Systems Center
Mechanical & Thermal Engineering Sciences Directorate
National Renewable Energy Laboratory (NREL), Golden, CO

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Transforming ENERGY
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NREL at-a-Glance

4,400

Workforce, including

205 postdoctoral researchers

179 graduate students

94 undergraduate students

World-class

facilities, renowned technology experts

More than 1000

Partnerships

with industry, academia, and government

Campus

operates as a living laboratory

Introduction

- B.S. Chemical Engineering, Washington University in St. Louis, 2012
- Ph.D. Chemical Engineering, Northwestern University, 2017; *doctoral research sponsored by Dow Chemical*
  - Dissertation Title: "Understanding Silica-Supported Group 4 and 5 Metal Oxide Catalysts for Selective Oxidations with Hydrogen Peroxide and for Epoxide Activation"
- Former graduate intern at 3M, Corporate Research Materials Laboratory, 2015; *adhesive formulation R&D*
- ***Senior Reaction Engineer (and other roles) at the National Renewable Energy Laboratory, 2017–present***
- Adjunct Professor of Chemical Engineering, Colorado School of Mines, January 2024–present

Overview of what I do for a living

I investigate **catalysts & reactors** for the **sustainable** production of **chemicals & fuels**.

"I make the stuff that makes the stuff that makes the stuff, **better**."

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Chemical engineering – chemistry… at the largest possible scale!

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Image from Google Images. Accessed 2017 April 17.

Chemical engineering – chemistry… at the largest possible scale!

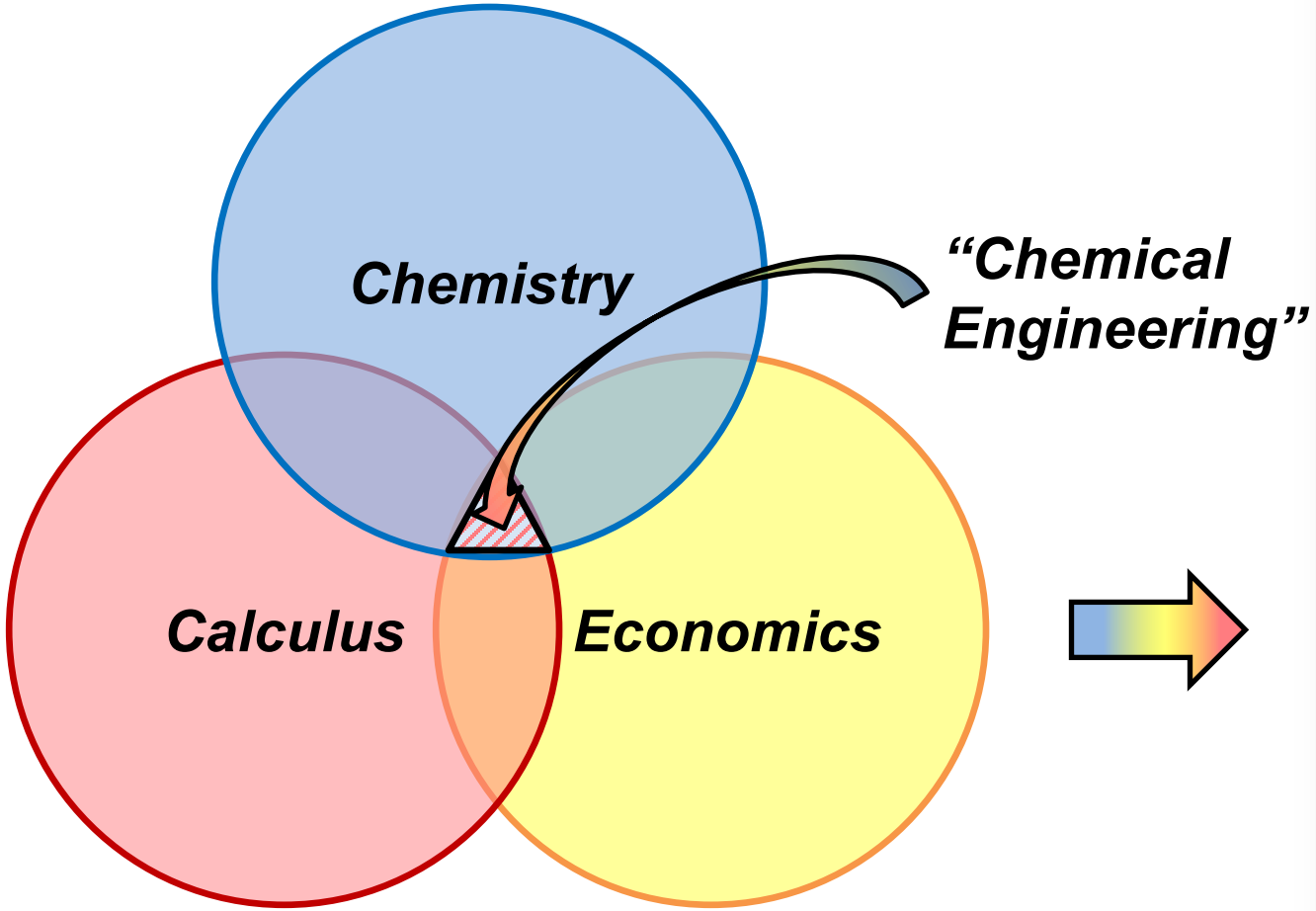

Image credit: Linda Wang, Chemical & Engineering News, Vol. 94 p.18, 28 June 2016. Accessed 17 April 2017.

Chemical engineering – chemistry… at the largest possible scale!

- **Industrial chemicals** are EVERYWHERE in our homes, cars and everyday lives
  - performance plastics & fabrics
  - foams, coatings, paints & adhesives
  - cosmetics, hygiene & cleaning products
  - pharmaceutical drugs
  - ➢ **nitrogen fertilizer (*today's lecture*)**

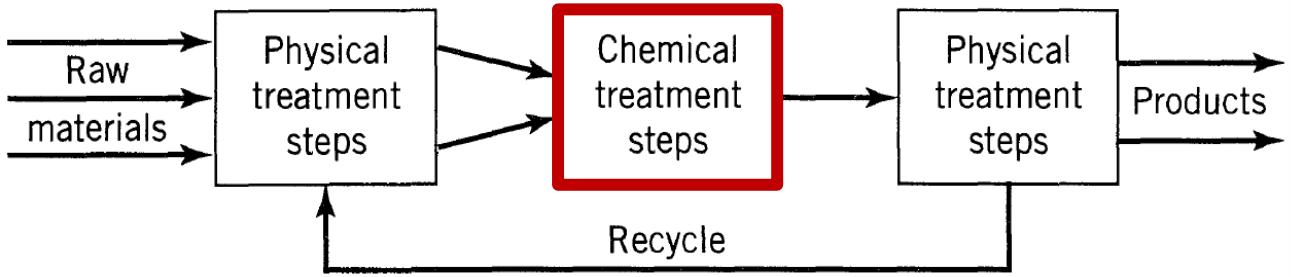

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Image credit: Linda Wang, Chemical & Engineering News, Vol. 94 p.18, 28 June 2016. Accessed 17 April 2017.
Levenspiel, O. Chemical Reaction Engineering, 3rd edition, Figure 1.1, p.1 (2003).

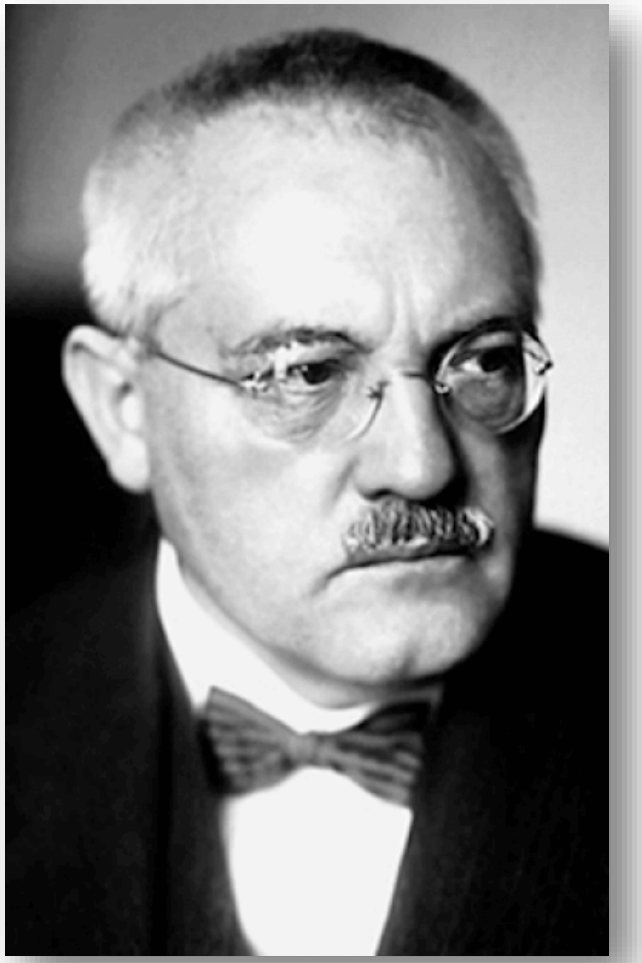
#TBT: the dawn of the 20th Century

Image from Google Images. Accessed 2023 December 4.

A tale of two scientists

Fritz Haber

Carl Bosch



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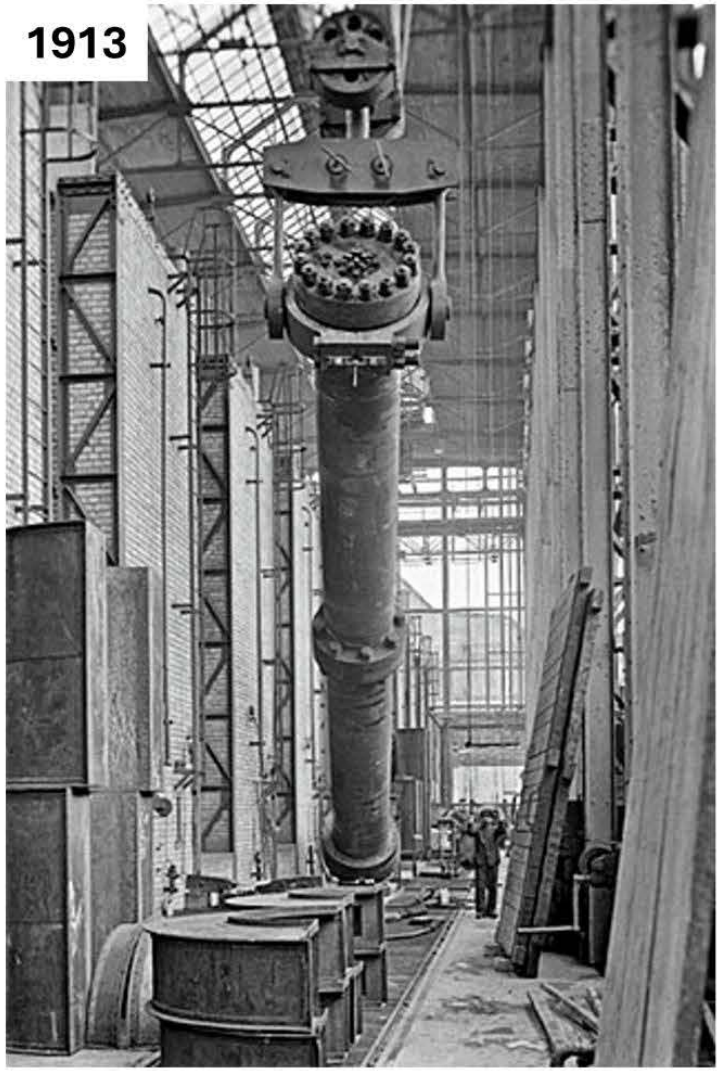
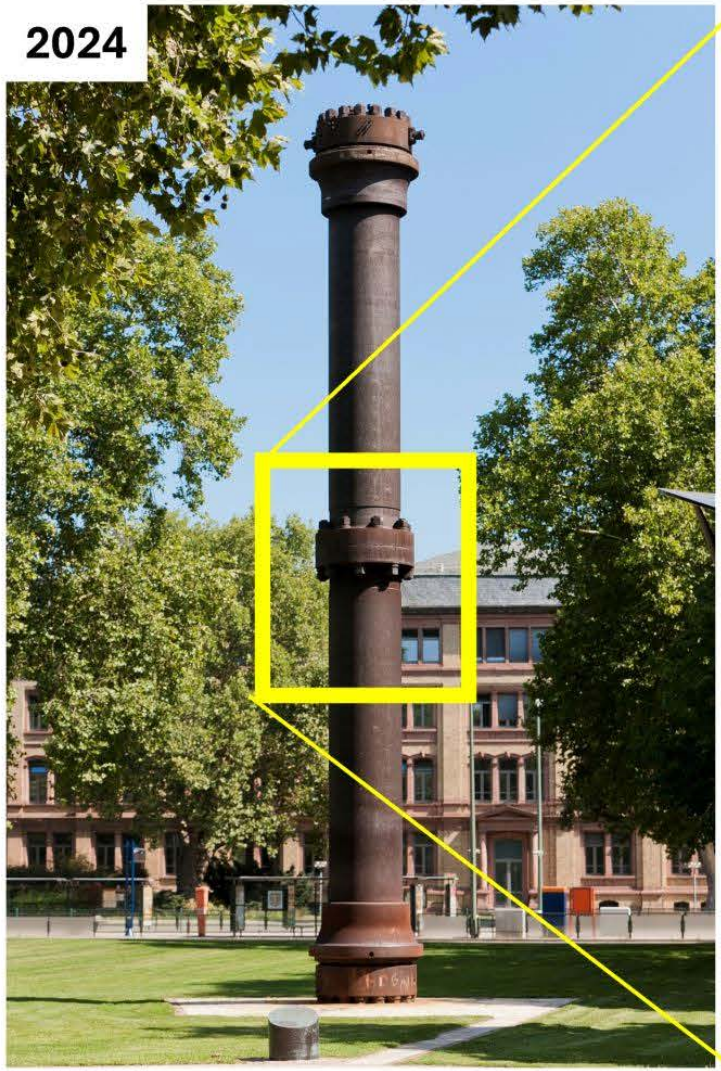
Images from Google Images. Accessed 2023 December 4.

111th anniversary of ammonia synthesis (and Carl Bosch's 150th birthday)





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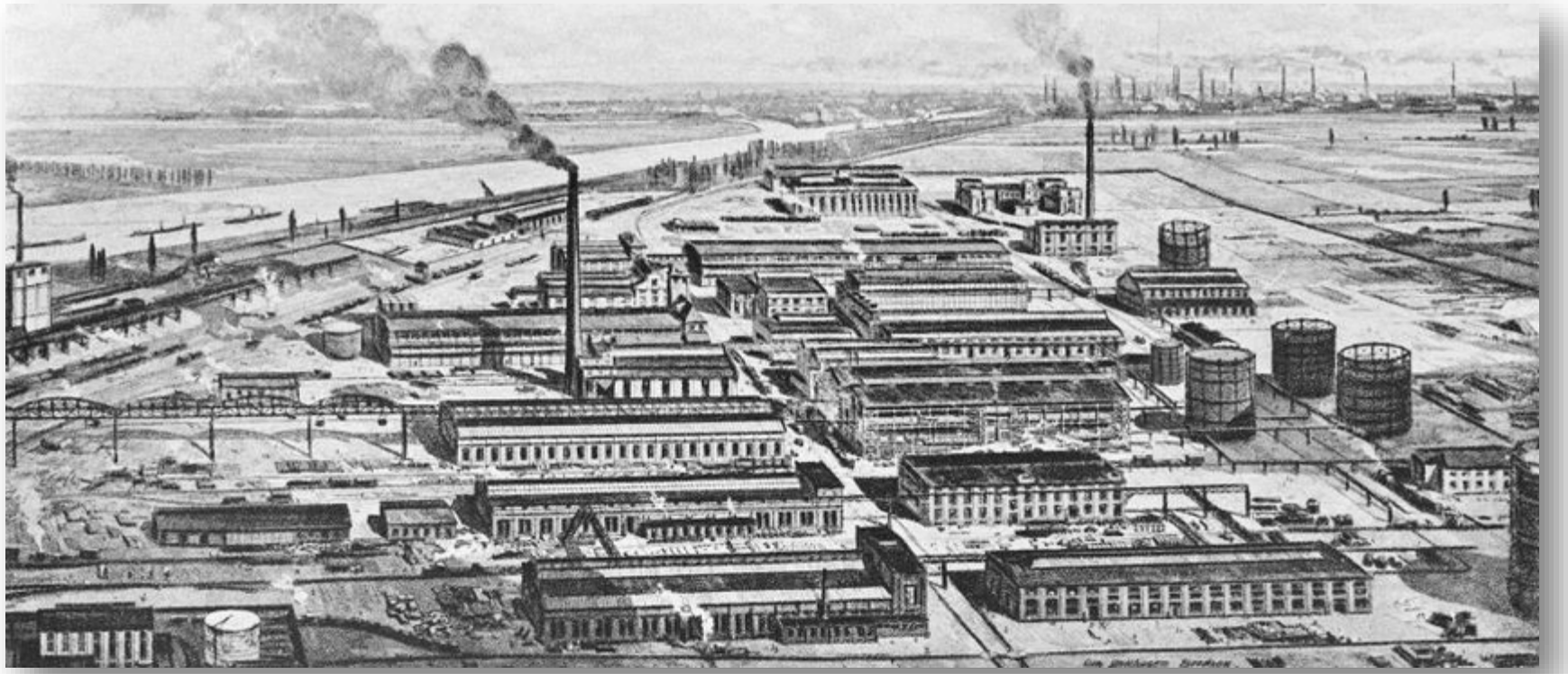
Images adapted from LinkedIn user post. Accessed 2024 September 23.

Life—and death—from ammonia's very discovery



BASF ammonia plant in Oppau (Ludwigshafen, Germany, 1913)

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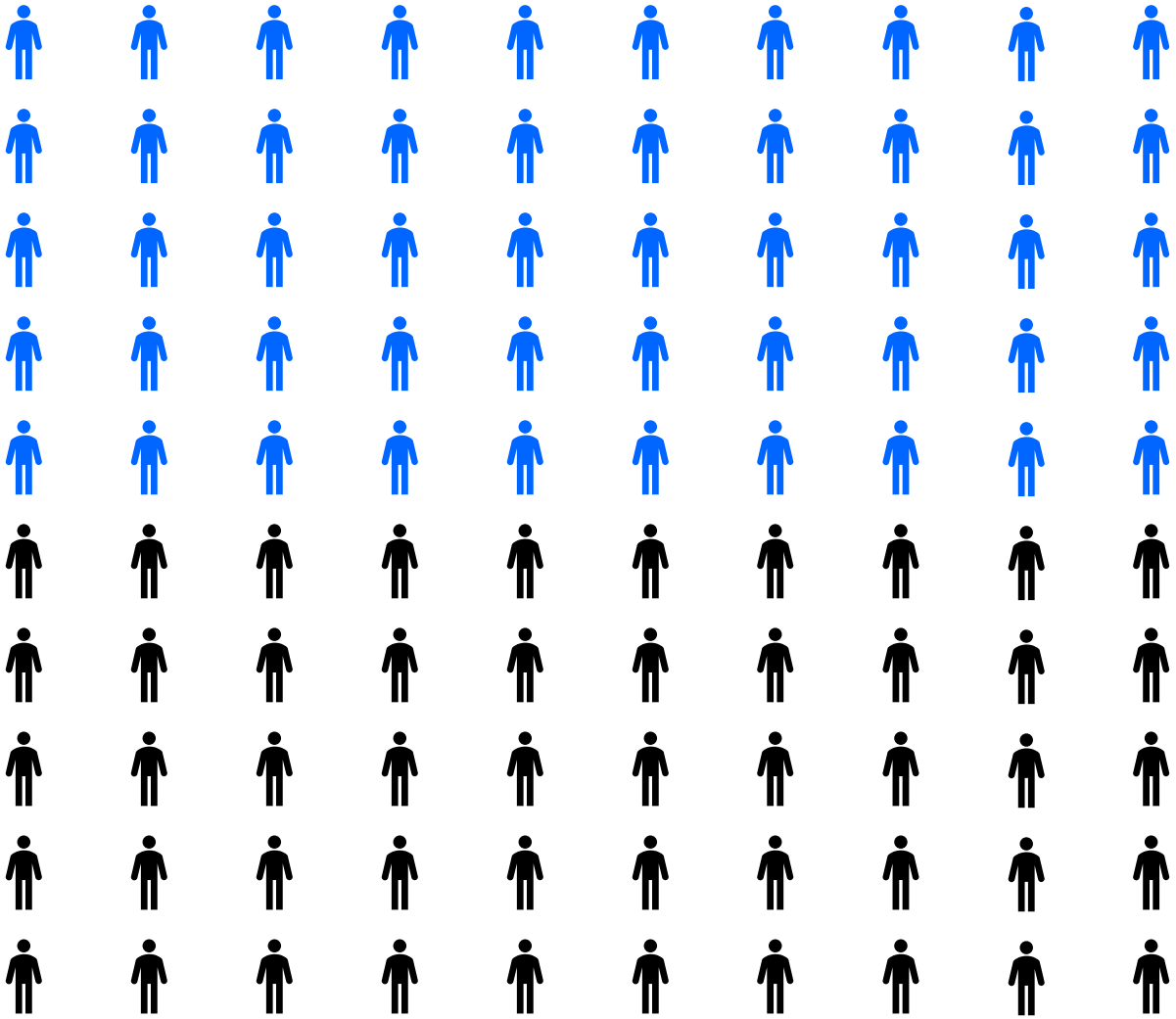
Images from Google Images. Accessed 2020 September 8.

50–50



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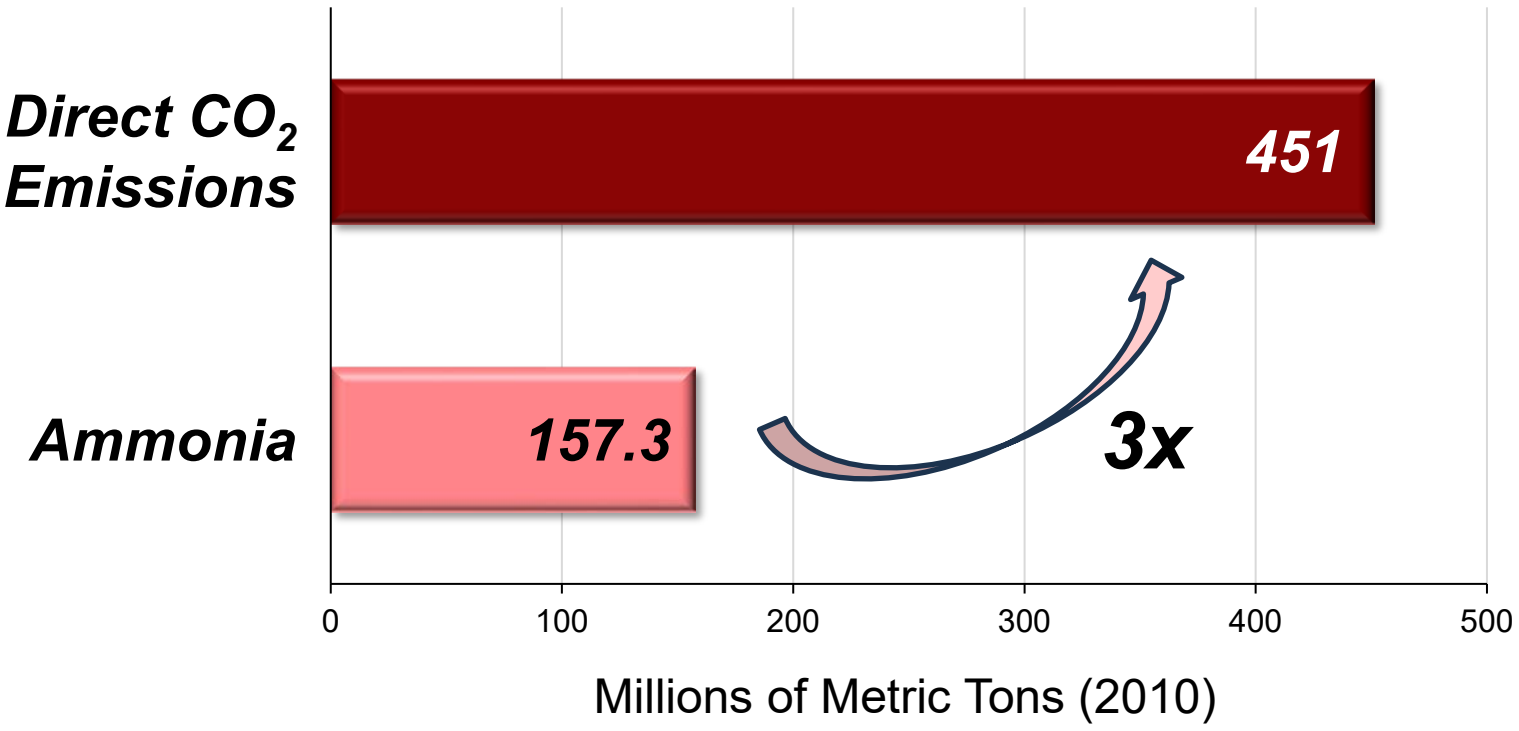
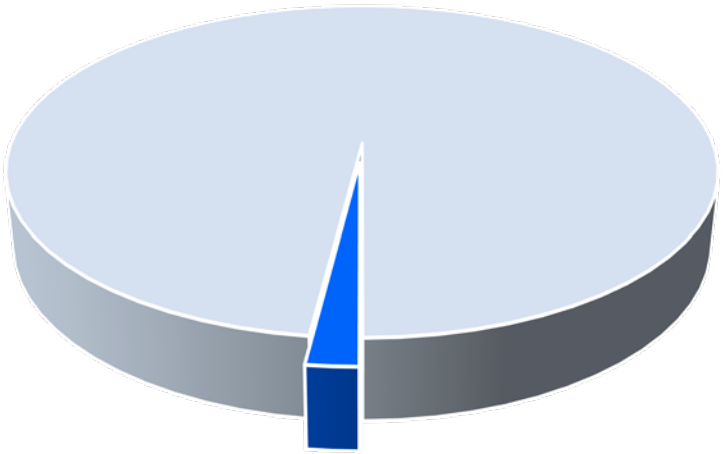
BUT… feeding the planet hurts the planet





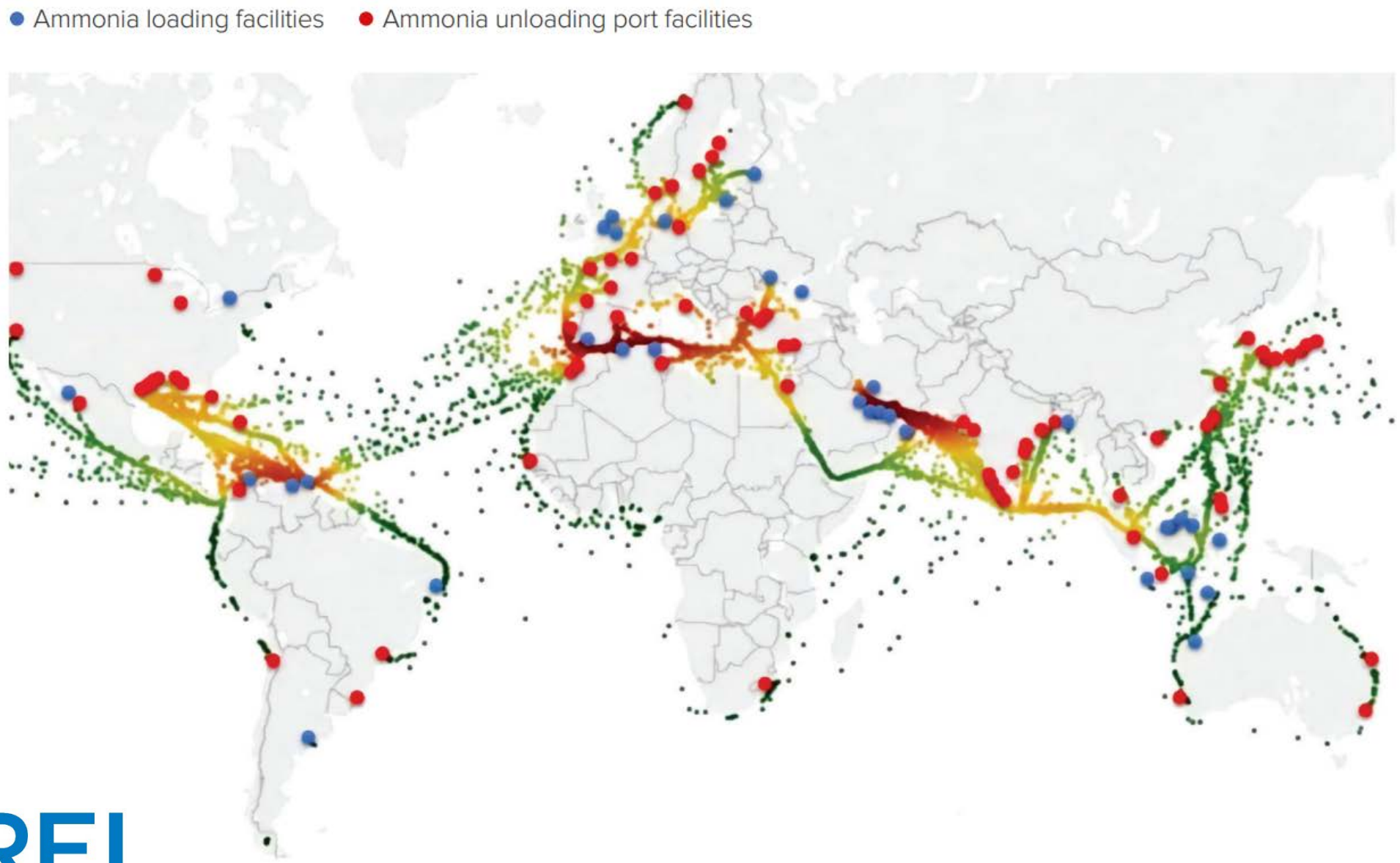
NH_3 consumes

1–2%

of the world's energy supply

Statistics from Institute for Industrial Productivity.

A crisis of food equity



Graphic adapted from The Royal Society's Ammonia Policy Briefing, 19 February 2020. Accessed 4 December 2023.

Rethinking the paradigm

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Images from Google Images. Accessed 2023 December 4.

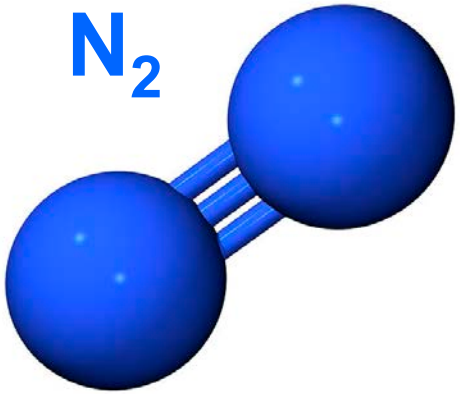
An uphill battle

$$N_2 + 3\,H_2 \rightleftharpoons 2\,NH_3$$

The First Life Lesson of Thermodynamics

You break it, you buy it.

N_2



Nitrogen's **triple bond** is extremely stable—***requiring <u>lots</u> of energy to break it.***

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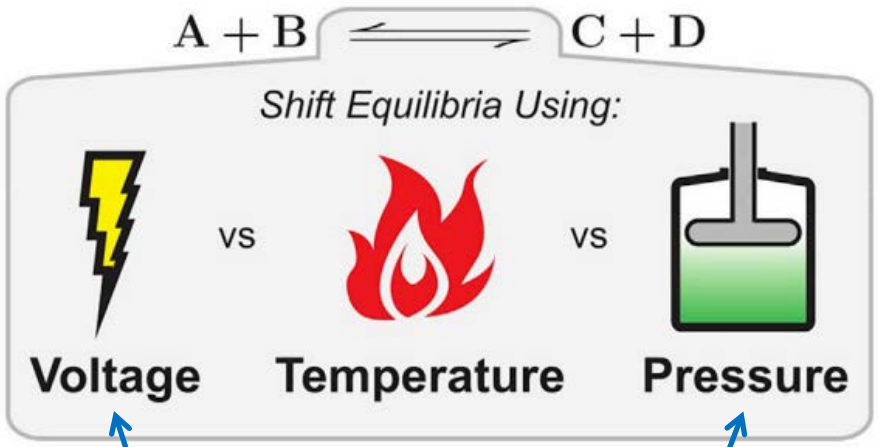
Statistics from global commodity chemical market reports.
Images from Google Images. Accessed 2020 September 8.

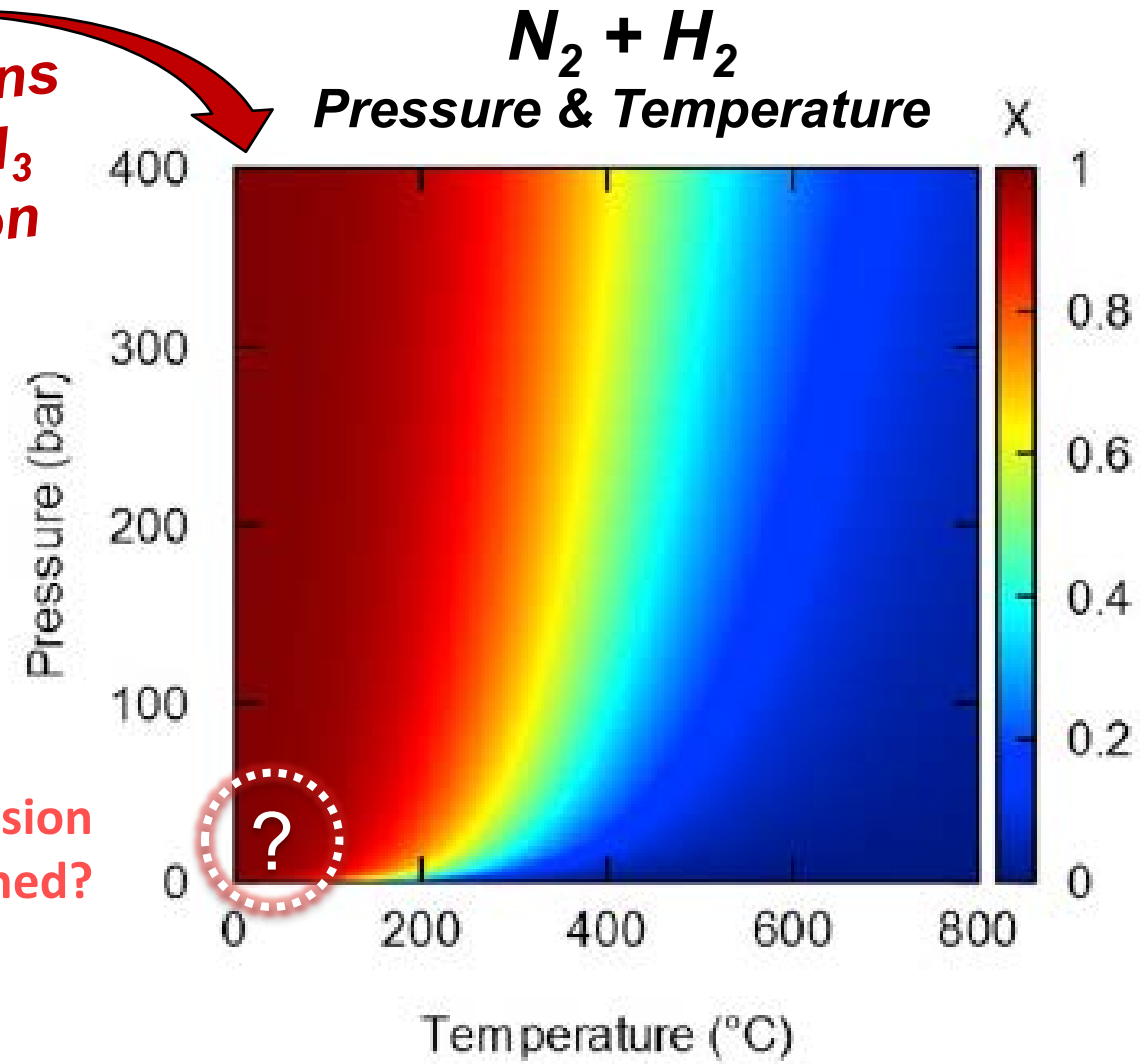
High school refresher: chemical reactions need heat—and a little push

$$N_2 + 3\,H_2 \rightleftharpoons 2\,NH_3$$



the "water pressure" that pushes chemical reactions

Red regions favor NH_3 formation



...mission accomplished?

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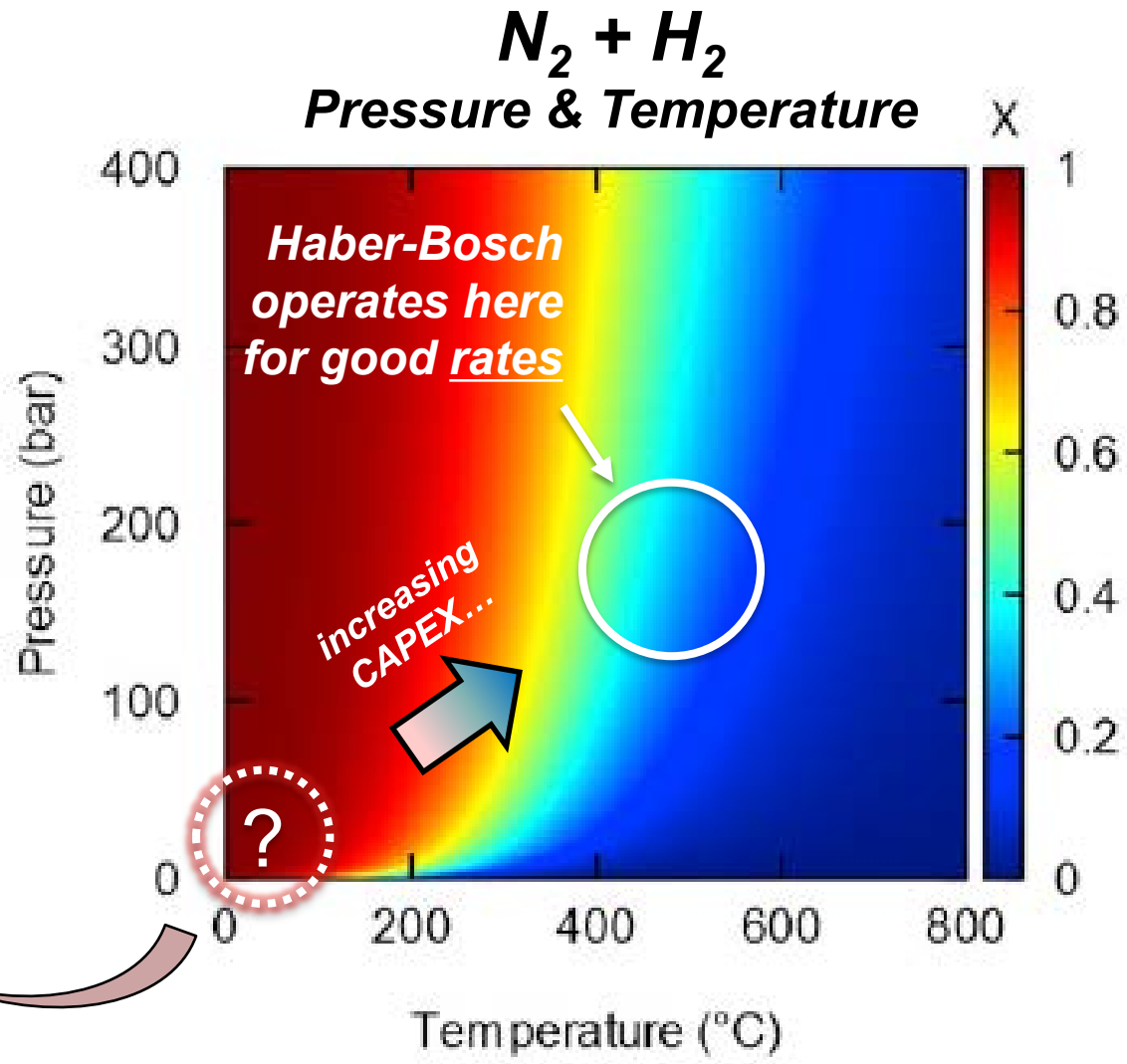
Figures adapted from: Z. J. Schiffer, K. Manthiram, *Joule* 1 10–14 (2017).

Thermodynamics are a real bugaboo to making chemistry happen fast

The Second Life Lesson of Thermodynamics

Just because the universe *really* wants something to happen… doesn't mean it will happen anytime soon.

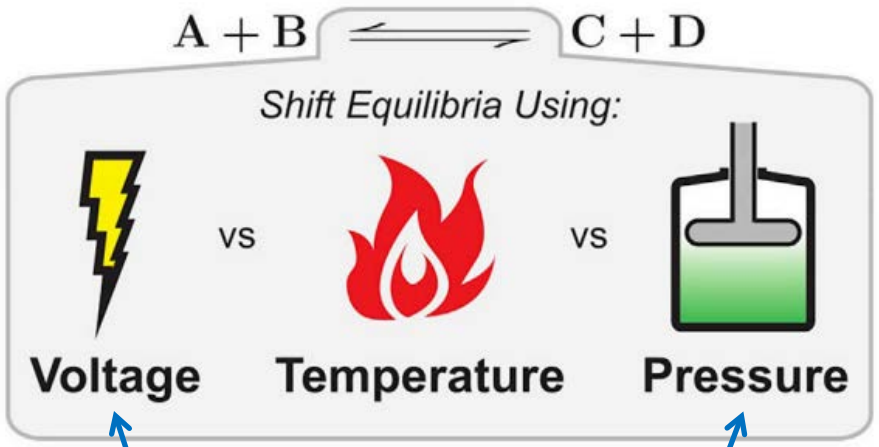
Chemical kinetics (rates) tell us how fast!



Figures adapted from: Z. J. Schiffer, K. Manthiram, *Joule* 1 10–14 (2017).

Could ***renewable electrons*** give ammonia synthesis a green "push"?

$$N_2 + 3\,H_2 \rightleftharpoons 2\,NH_3$$

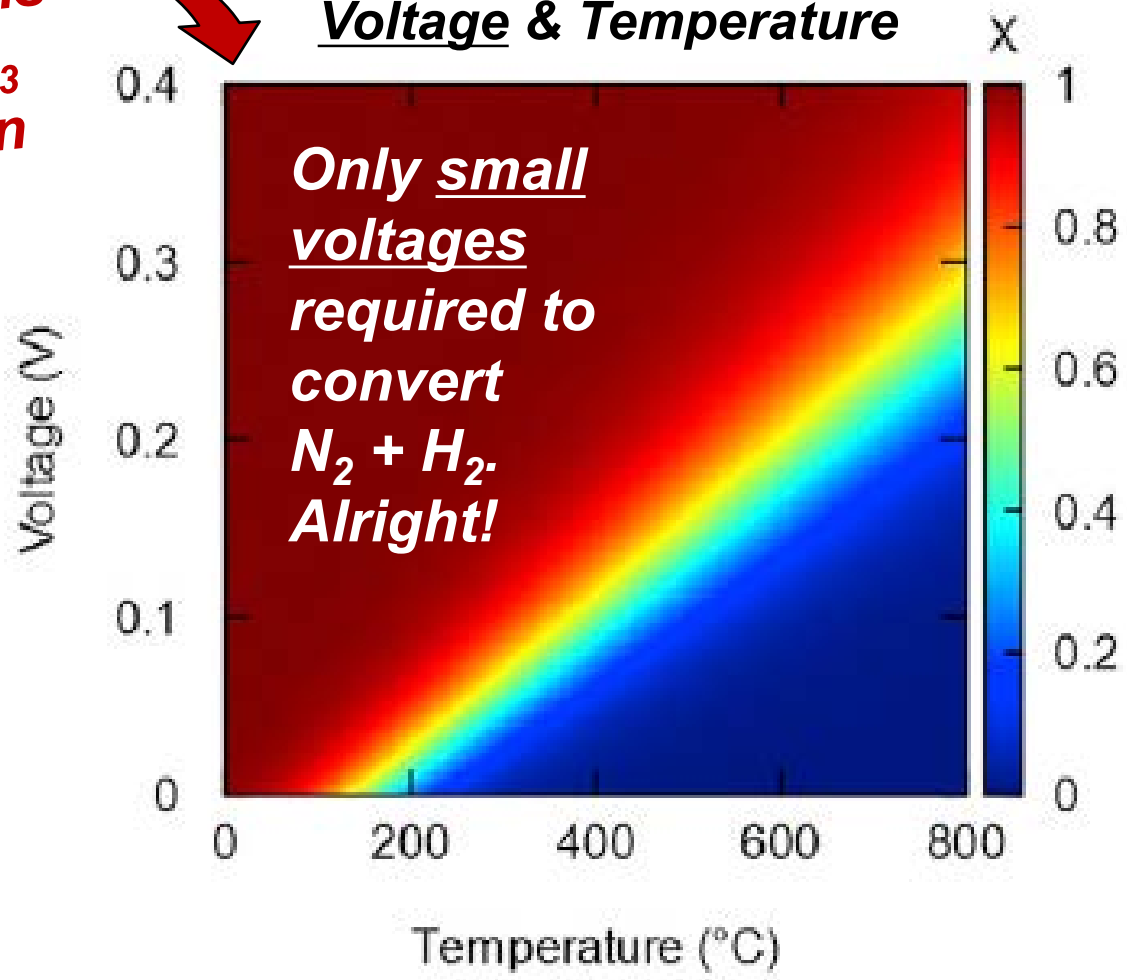


the "water pressure" that pushes chemical reactions

Red regions favor NH_3 formation

$N_2 + H_2$

Voltage & Temperature



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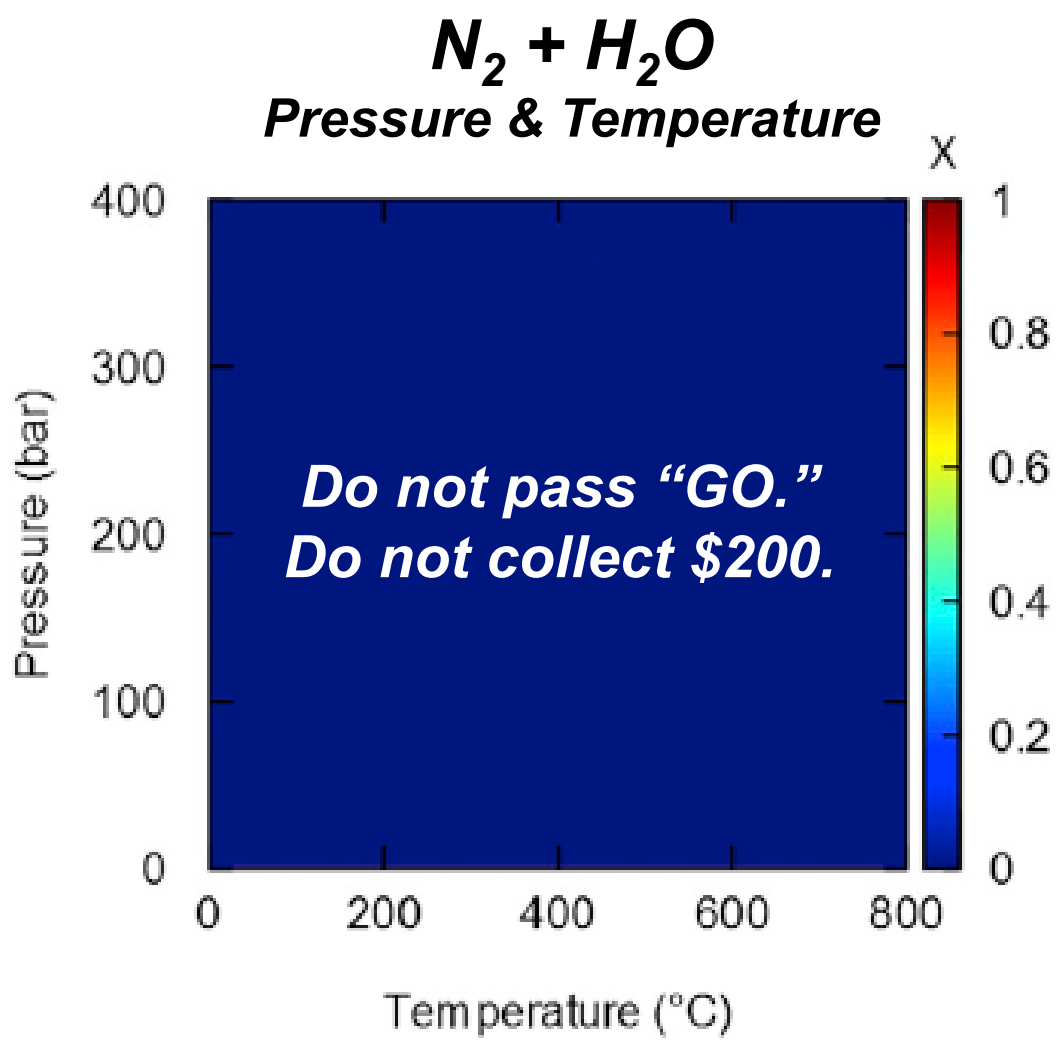
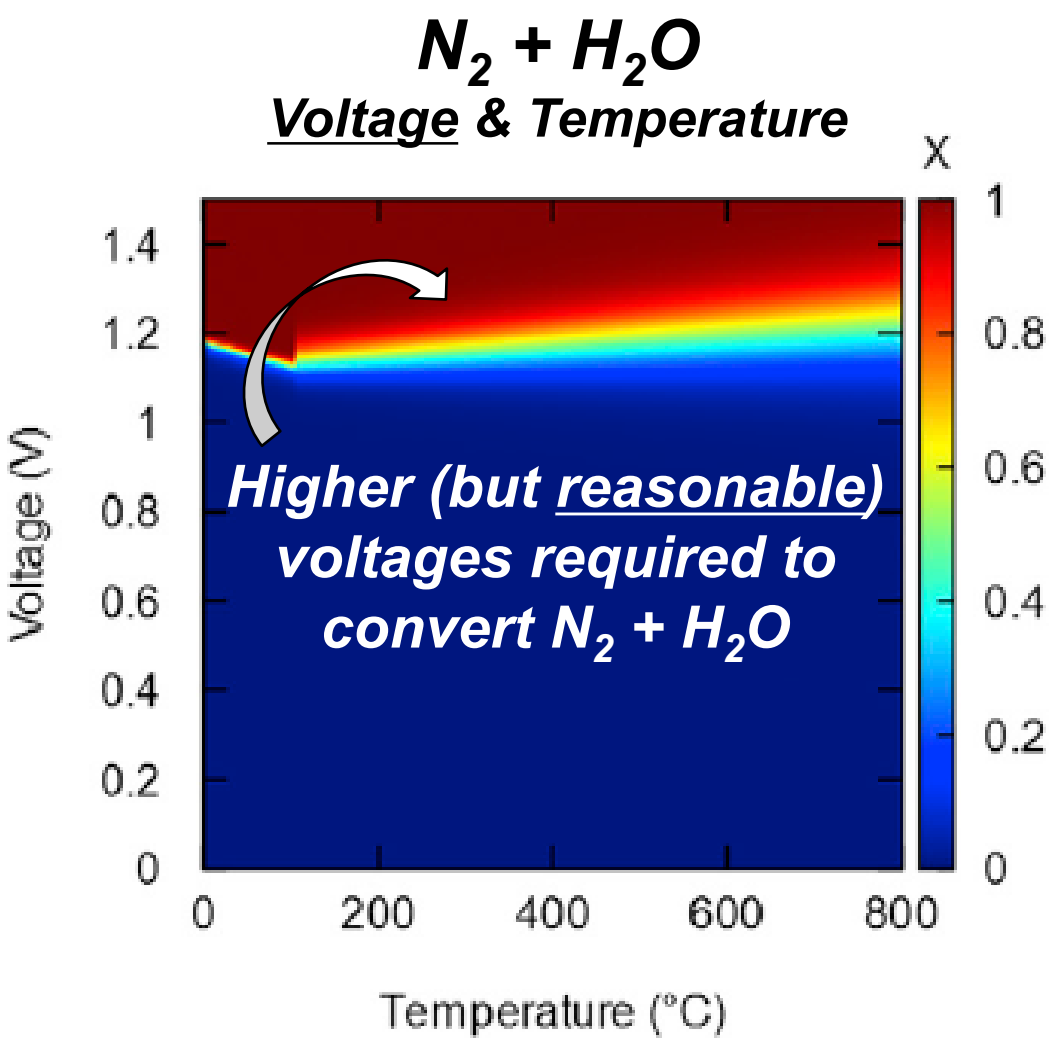
Figures adapted from: Z. J. Schiffer, K. Manthiram, *Joule* 1 10–14 (2017).

What happens when we try to use water (H_2O) as our source of hydrogen?





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Figures adapted from: Z. J. Schiffer, K. Manthiram, *Joule* 1 10–14 (2017).

Electrochemistry unlocks possibilities to make ammonia using renewables

Electrochemistry helps tip the scales in ammonia's favor and allows us to utilize renewable inputs.

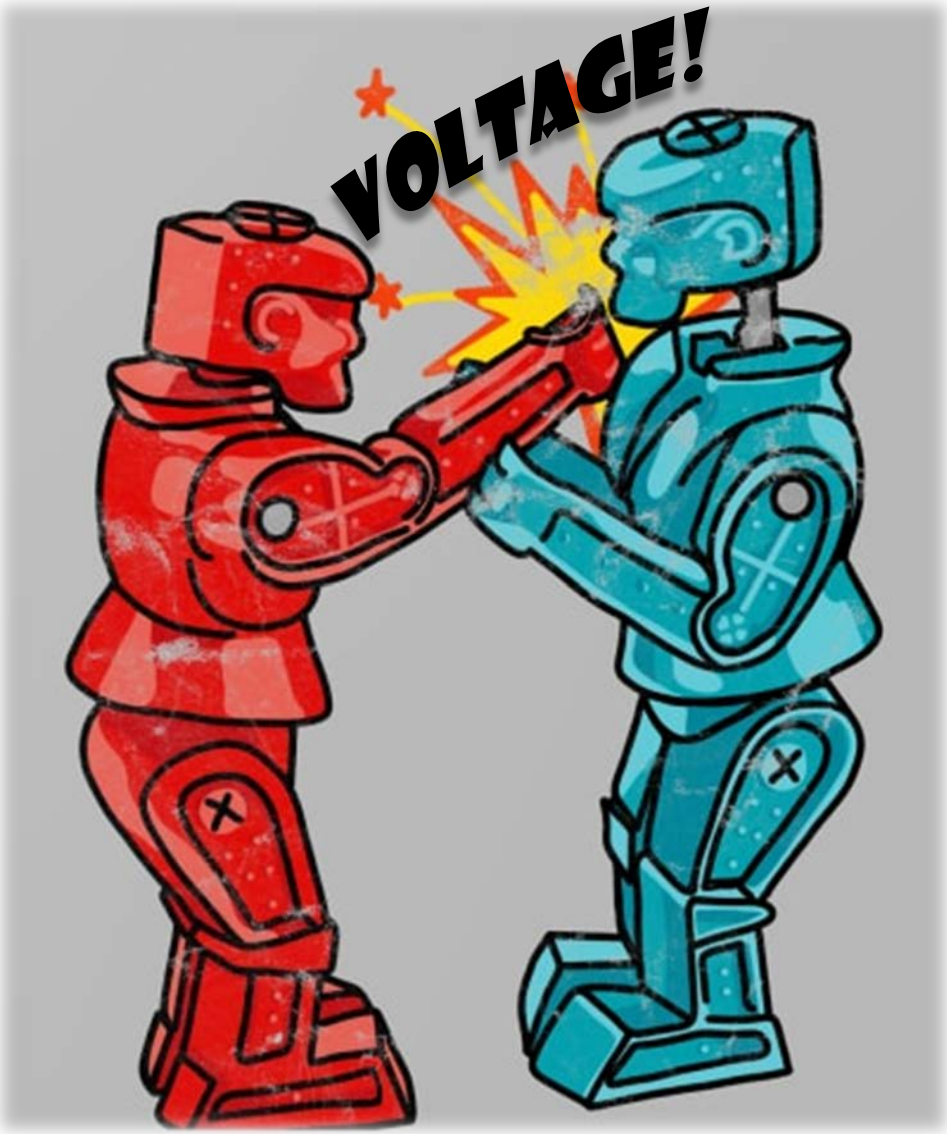
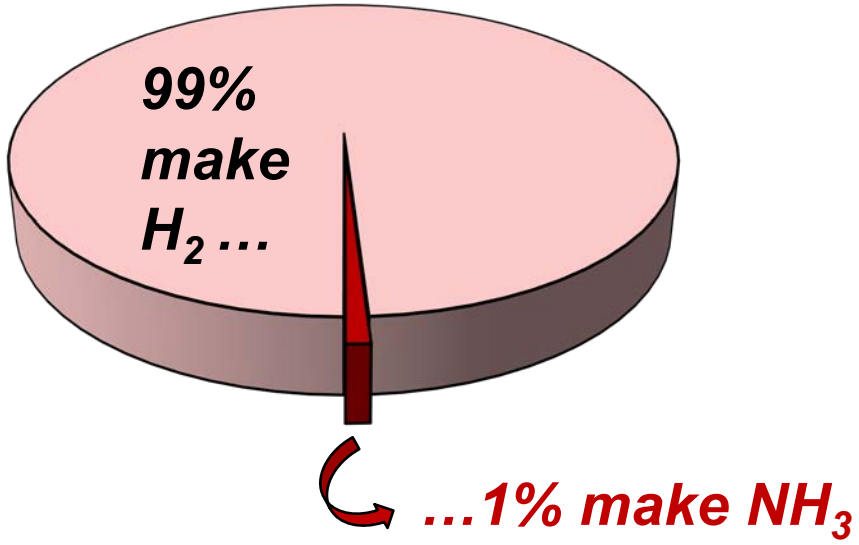


Image from Google Images. Accessed 2023 November 14.

What to expect when you're ~~expecting~~ electrifying

The Third Life Lesson of Thermodynamics

Electrons are a lot like kids.



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Modularization unlocks the future of renewable NH_3 and food equity

Creates fertilizer <u>when</u> and <u>where</u> we need it…

…only needs air, water and renewable electricity

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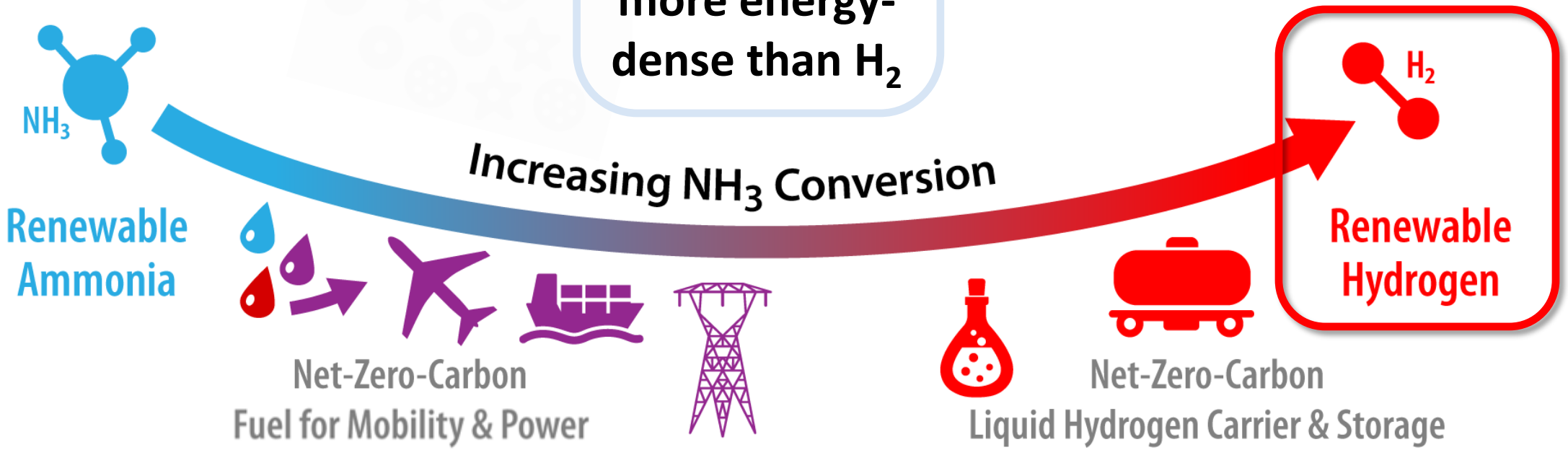
Image adapted from M2X Energy, Inc. Accessed 4 December 2023.

Reversing Haber-Bosch

Liquid NH_3 is

1.5x

more energy-dense than H_2



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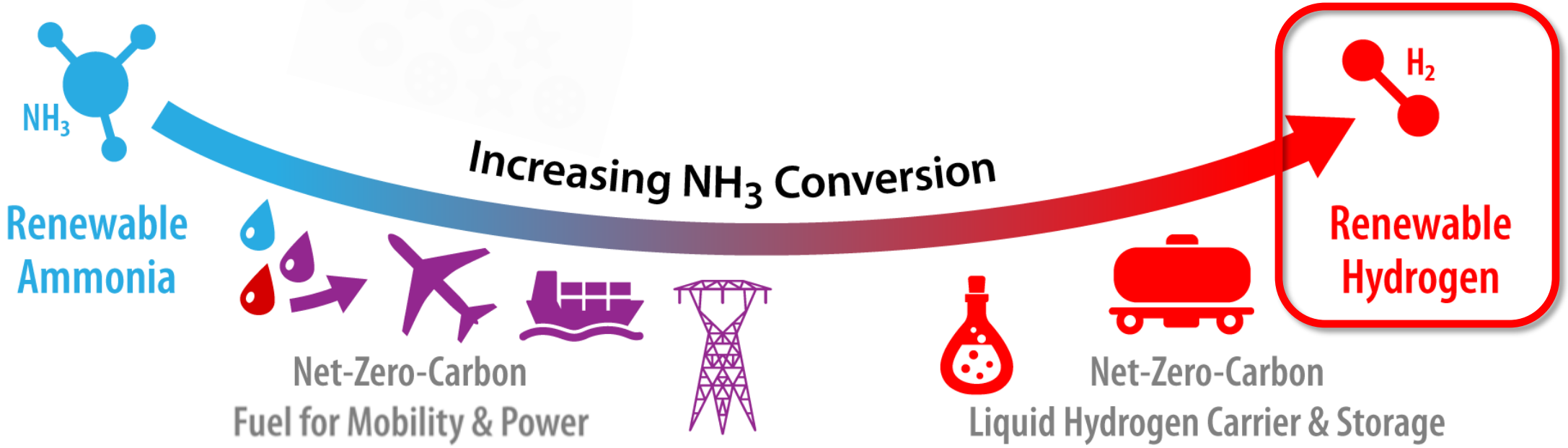
Graphic adapted from L. C. Caballero, N. E. Thornburg, M. M. Nigra, *Chemical Science* 13 12945–12956 (2022)

Reversing Haber-Bosch

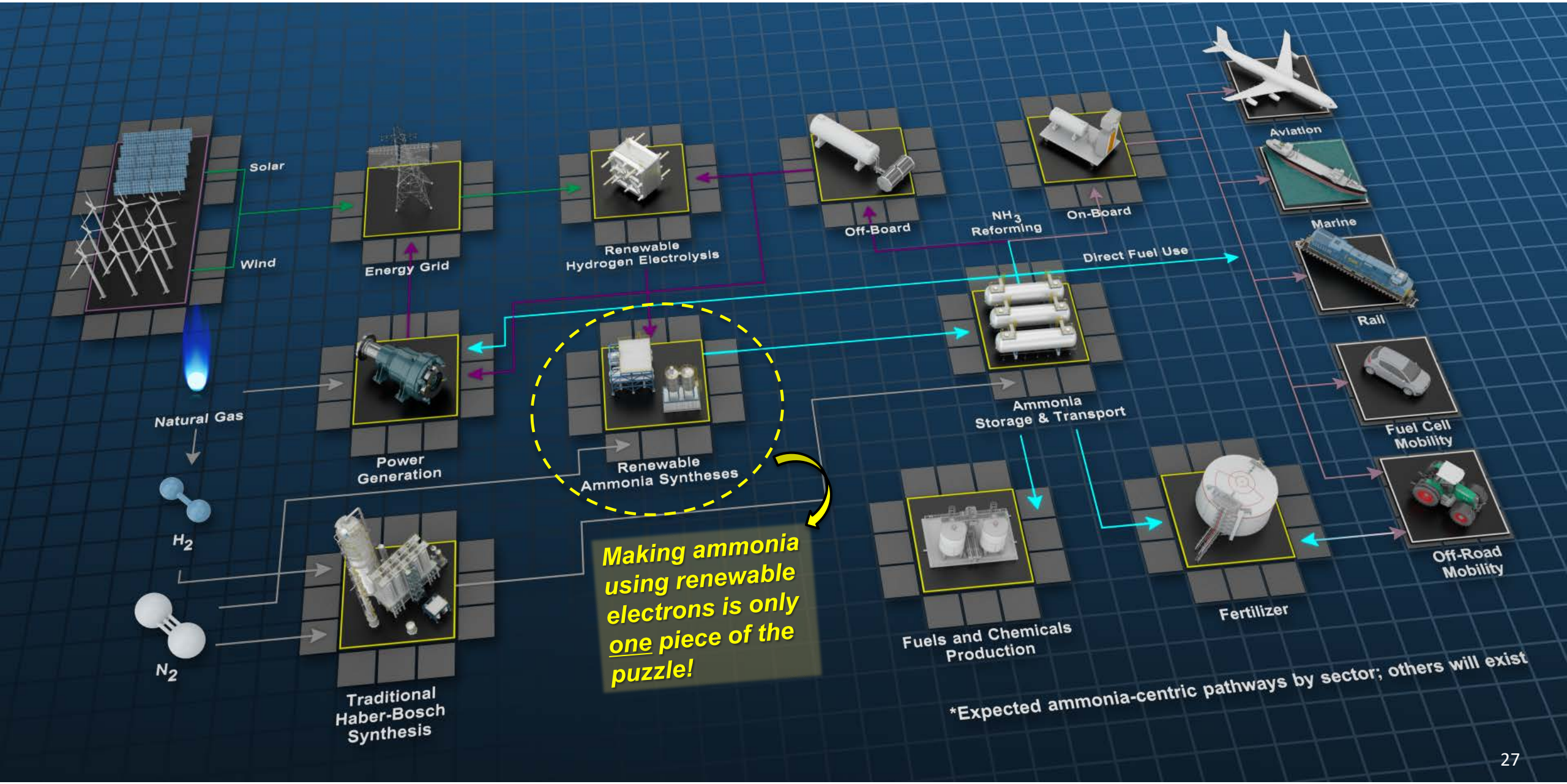
The Fourth Life Lesson of Thermodynamics

Sometimes, things work out the way you want.

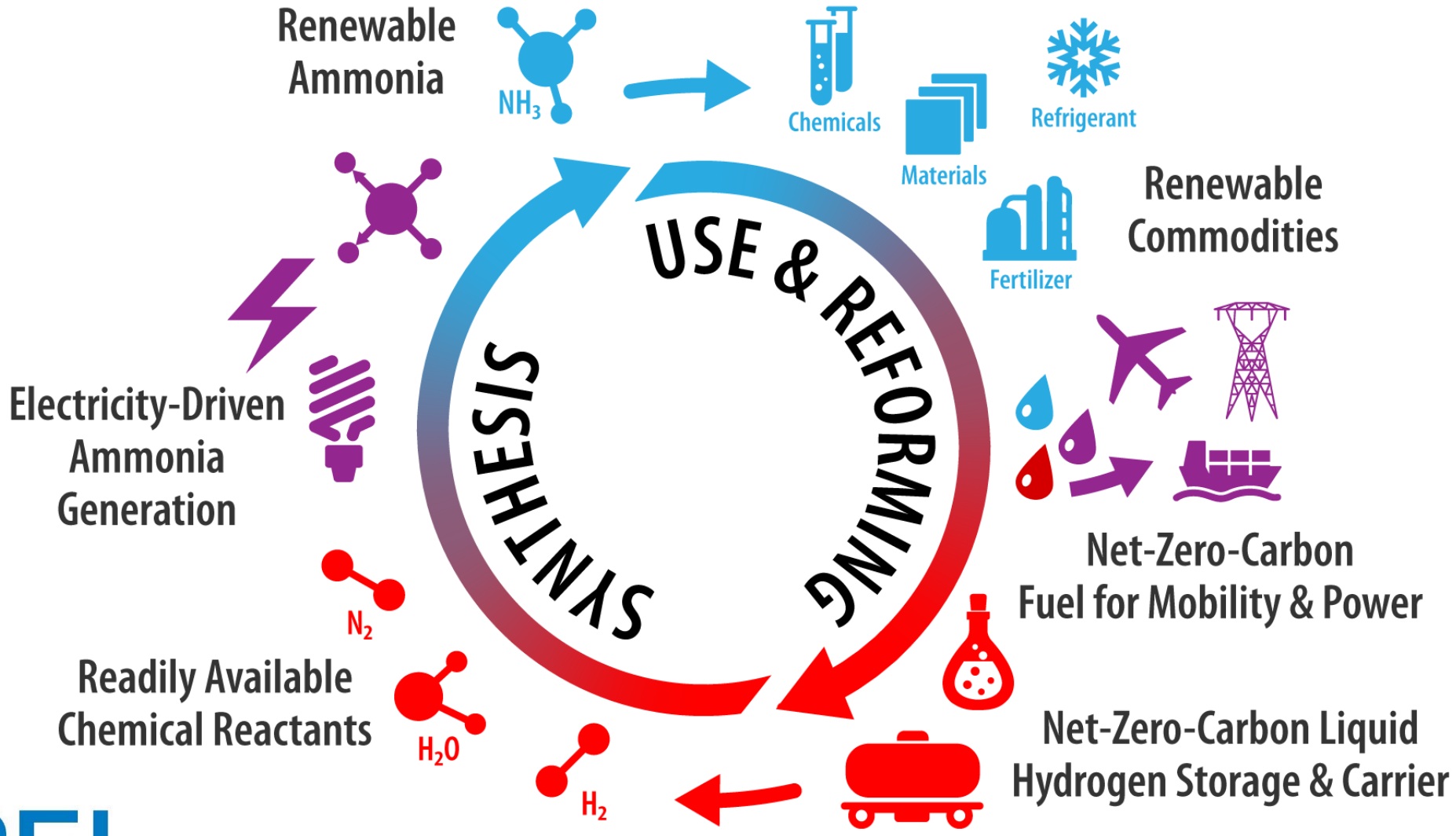


Graphic adapted from L. C. Caballero, N. E. Thornburg, M. M. Nigra, *Chemical Science* 13 12945–12956 (2022)

NREL is helping build the future ammonia energy economy



Circularity of nitrogen and hydrogen



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Thank You

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